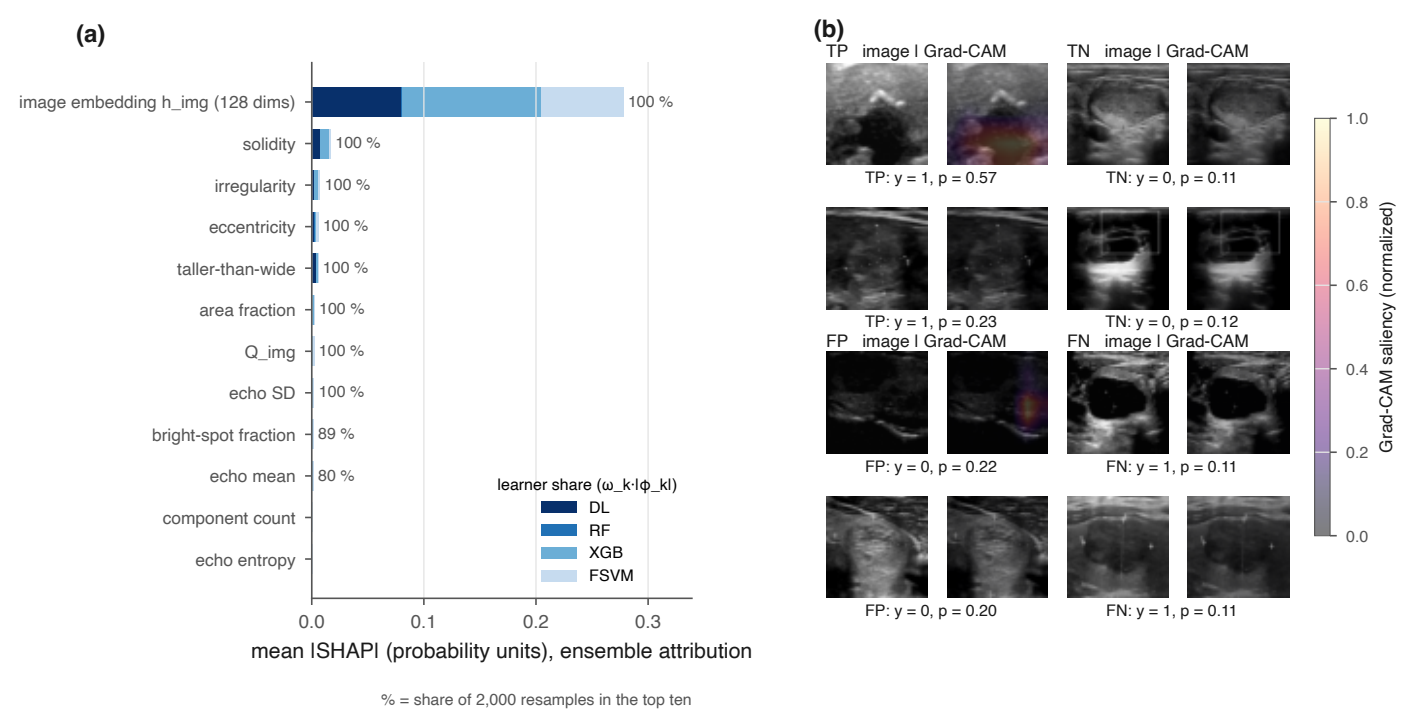

(a)
image embedding h_img (128 dims)
solidity
irregularity
eccentricity
taller-than-wide
area fraction
Q_img
echo SD
bright-spot fraction
echo mean
component count
echo entropy
100 %
100 %
100 %
100 %
100 %
100 %
100 %
89 %
80 %
learner share (ω_k·lφ_kl)
DL
RF
XGB
FSVM
0.0
0.1
0.2
0.3
mean ISHAPI (probability units), ensemble attribution
% = share of 2,000 resamples in the top ten
(b)
TP image | Grad-CAM
TN image | Grad-CAM
TP: y = 1, p = 0.57
TN: y = 0, p = 0.11
TP: y = 1, p = 0.23
TN: y = 0, p = 0.12
FP image | Grad-CAM
FN image | Grad-CAM
FP: y = 0, p = 0.22
FN: y = 1, p = 0.11
FP: y = 0, p = 0.20
FN: y = 1, p = 0.11
Grad-CAM saliency (normalized)
1.0
0.8
0.6
0.4
0.2
0.0